

Cybersecurity (F26)

4th February 2026

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Welcome
to a new
semester

Teaching Staffs

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Philosophy

- ✓ Read at Home
- ✓ Work in Class
- ✓ Discuss in Groups
- ✓ *You are responsible for your own learning, and your own **action***

Lecture Structure

- ✓ **Venue:** Sønderborg ACL Løkken 136
- ✓ **Lecture** [12:30 – 14:30]- [Every Wednesday](#)
- ✓ **Demo** after each lecture/during lecture (depending on time availability)
- ✓ **Lab Exercises** [14:30 – 16:15]
- ✓ Course is revised continuously
- ✓ Slides will be adapted after the lectures
- ✓ Cybersecurity incident of the week: (Attack/threat incident in the previous week(s), month(s), or year(s) - its cause, and possible mitigation strategy)**

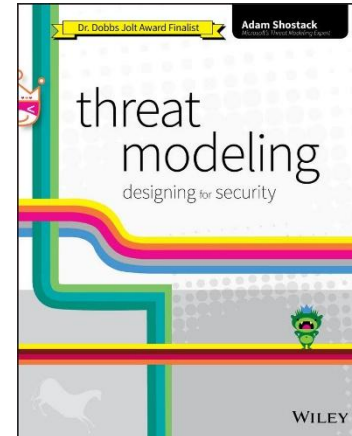
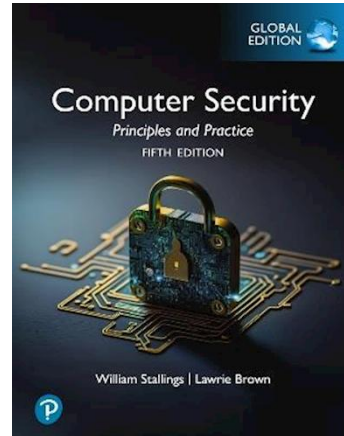
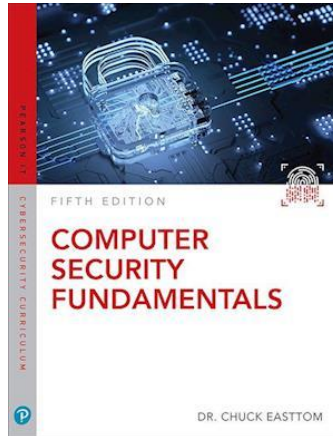
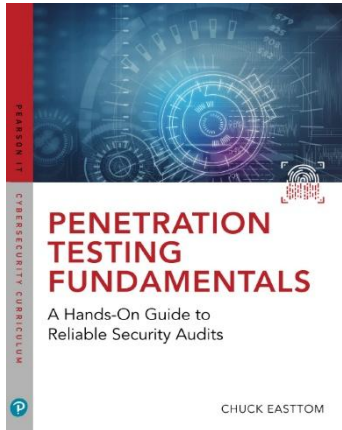
Course Plan

Lecture	Date	Teacher	Topic
01	2026-02-04	Sani	Introduction & Lab Set-up
02	2026-02-11	Sani	Vulnerability Assessment
03	2026-02-18	Sani	Penetration Testing
04	2026-02-25	Hiraku	Firewalls and IDS
05	2026-03-04	Hiraku	Malware & SQL
06	2026-03-11	Hiraku	Denial of Service
07	2026-03-18	Sani	Social Engineering, Phishing
08	2026-03-25	Sani	DDoS, MITM
09	2026-04-08	Hiraku	Cryptography
10	2026-04-15	Hiraku	Multi-party Computation
11	2026-04-22	Hiraku	Privacy-preserving Data Analysis
12	2026-04-29	Sani	Theat Modelling

Curriculum

✓ Primary Literature (Books)

- Computer Security Fundamentals by Chuck Easttom, 5th Edition
- Penetration Testing Fundamentals: A Hands-on Guide to Reliable Security Audits by Chuck Easttom, 4th Edition
- Threat Modeling: Designing for Security by Adam Shostack
- Computer Security: Principles and Practice, 5th Edition, by William Stallings and Lawrie Brown



✓ Secondary Literature

- Online Resources
- Articles
- Blog Posts
- Videos

Knowledge Repositories (Not Exhaustive!)

- [Common Vulnerabilities and Exposures \(CVE\)](#)
- [Exploit Database](#)
- [Common Weakness Enumeration](#)
- [MITRE ATT&CK Framework](#)
- [Open Web Application Security Project \(OWASP\)](#)
- [CVE Details](#)

✓ Online Training Platforms

- [TryHackMe](#)
- [Hack The Box](#)
- [OverTheWire](#)

Tools

✓ Penetration testing

- Nmap
- Netcat
- Metasploit Framework
- Burp Suit
- Kali Linux Toolset
- WireShark
- Scapy (mostly advance testing & research)
- Social Engineer Toolset

✓ Vulnerability Assessment

- Nessus
- OpenVas / Greenbone
- Rapid7 InsightVM (formerly Nexpose)

✓ Virtualization Platforms

- VirtualBox / VMware

✓ Vulnerable Machines

- Metasploitable 2 & Metasploitable 3

✓ Firewall/IDS/IPS

- Snort

✓ SIEM (Inc. Extended Detection & Response)

- Wazuh

Ethical Concerns

- Participants must agree to uphold ethical standards in Cybersecurity.
- Knowledge and skills gained are solely for educational purposes.
- Any unauthorized access, manipulation, or misuse of data or systems is prohibited.
- You are responsible for your own actions.
- Sign the **Ethical Statement** Form in ItsLearning

Exam and Exercises

- ✓ Written Exam based on 6 Selected Hands-in Exercises
- ✓ The 6 Hands-in Exercises Serve as Materials for the final Exam.
 - Combining MCQ with Short Essay Questions (SEQ)
- ✓ Form a group for the exercises (groups of 4-6 welcome)
- ✓ Questions in English
- ✓ 120 minutes
- ✓ Grading [-3 — 12]

Expectation

- ✓ Participation and Engagement
- ✓ Exercises with 6 hand-ins
- ✓ Support and Communication Channels (e.g., Course Discord)
- ✓ **Goal:** Built a Strong **Purple Teams** of Cybersecurity Experts

Summary

- ✓ Lectures and Labs; Wednesday 12:30 – 16:15
- ✓ You are responsible for your own learning and ‘action’
- ✓ Six hands-in from exercises
- ✓ Written exam with MCQ, MCA and Essay
- ✓ Ethical concerns (sign the ethical statement)
- ✓ Check out the plan on ItsLearning

ItsLearning Plan

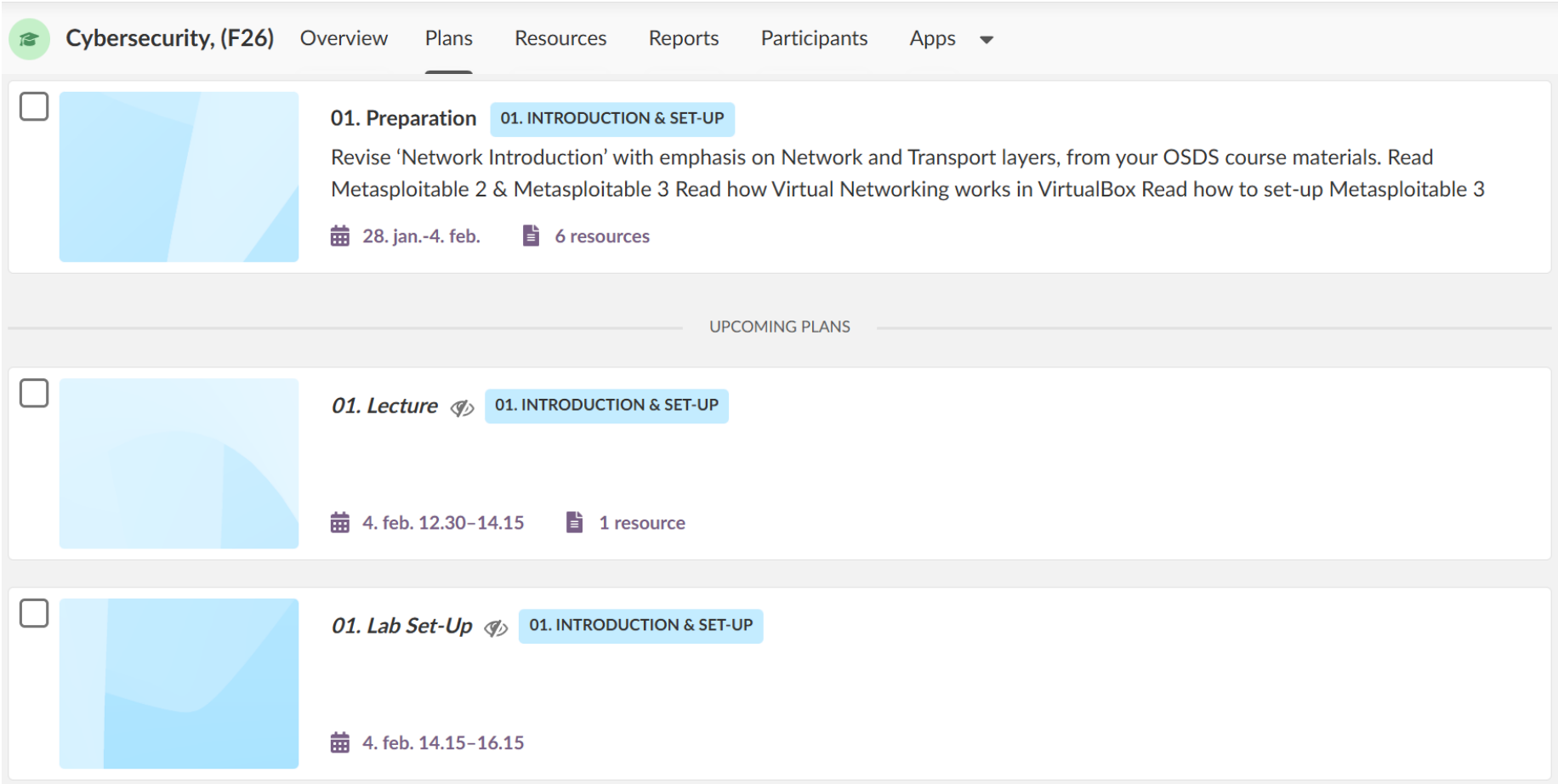


Figure: Screenshot from Cybersecurity course plan on ItsLearning

Introduction

Regarding drones over Denmark

UPDATED: The Danish Defence Command supports the Police with relevant capabilities in connection with drone observations in the vicinity of critical infrastructure and military installations.

September 27, 2025 - 11:30

B B C

Denmark reports more drone sightings ahead of European summits

Drones seen over Danish military bases in latest air disruption

27 September 2025

"Drones do not only represent the warfare of the future. They are a crucial part of our present".



The ban on civilian unmanned flights will last until Friday and comes as heads of EU governments gather amid spate of drone incursions



22 Energy Firms Hacked in Largest Coordinated Attack on Denmark's Critical Infrastructure

Denmark's SektorCERT association shares details on a coordinated attack against the country's energy sector.

news.cision.com / Formpipe Software / Cybersecurity incident at Formpipe's...

Cybersecurity incident at Formpipe's Danish subsidiary



FRI, OCT 18, 2024 14:00 CET
The Danish subsidiary of
As of now, Formpipe ass

Danish water infrastructure at 'very high risk' from cyber crime

- Formpipe has ta and business cc
- Formpipe has n Protection Agen
- The incident is i other internation
- The incident onl affected, meanir contacted and p
- The affected sys support service
- The financial im



Ritzau/The Local - news@thelocal.dk
Published: 22 Jan, 2025 CET. Updated: Wed 22 Jan 2025 12:41 CET

Save Add a comment f t in



Danish hosting firms lose all customer data in ransomware attack

CloudNordic and AzeroCloud have lost everything

August 24, 2023 By: Dan Swinhoe Have your say



Danish cloud hosting providers CloudNordic and AzeroCloud have lost access to all customers after a ransomware attack.

"Unfortunately, during the night of Friday 18-8-2023 at 04:00, CloudNordic/AzeroCloud was exposed to a ransomware attack, where criminal hackers shut down all systems," the companies said in matching statements on their respective websites. "Websites, e-mail systems, customer systems, our customers' websites, etc. Everything. A break-in that has paralyzed

Where are we getting it wrong?

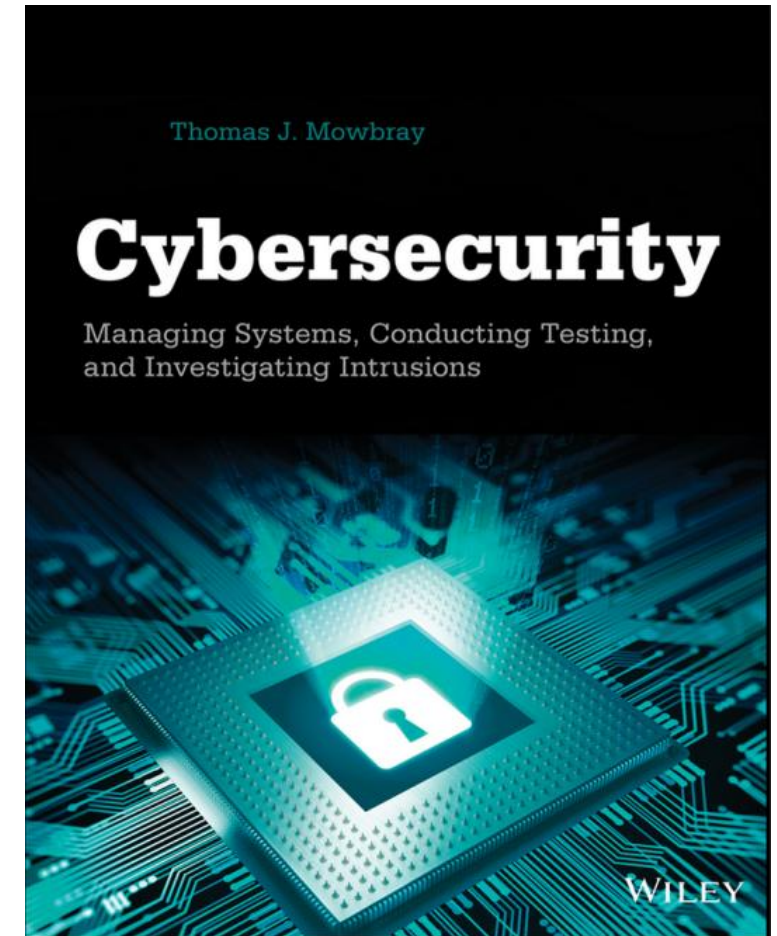
- Cyber criminals are responsible for about 11.36 trillion USD per year in financial losses, business downtime, and other assets.

[[“83 Cybersecurity Statistics 2026: Worldwide Data & Trends”](#)]

From the perspective of software engineer

- “The cybersecurity crisis is a fundamental failure of architecture.”
- Security should be—but is often not—considered during early software design.

[Mowbray, T. J., “[Cybersecurity: Managing Systems, conducting Testing, and Investigating Intrusions](#)”]



Where are we getting it wrong?

Many security issues can be either accidental or intentional

- Simple bugs in software and firmware
- Mis-configuration across systems and services
- Unintended side effects caused by system complexity (interconnectedness)
- Human factors (from unintentional errors to malicious actions)

How can we get to the top of it?

Possible ways to counter these issues include

- Best practices & security policies
- Checks for known bad patterns
- Defense-in-depth
- Secure by design principles
- Training and awareness
- Experience and continues learning



How can we get to the top of it?

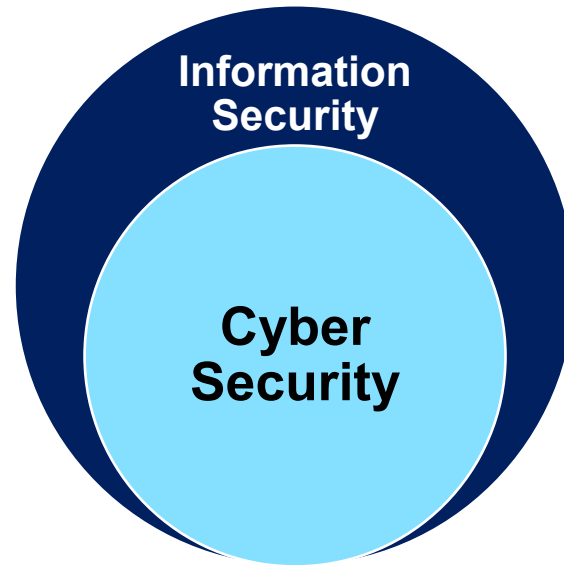
Understand Cybersecurity: Introducing risks & defense mechanisms

Major topics:

- **Vulnerability Assessment:** Finding weaknesses
- **Penetration Testing:** Ethical hacking to find exploitable weaknesses
- **Threat Modelling:** Understanding what can go wrong, how it can be attacked and what to do about it.
- **Security Monitoring:** SIEM, firewalls, intrusion detection, intrusion prevention, etc.
- **Incident Response:** Handling breaches when they occur
- **Hardening:** Improving system security – two-factor authentication, patches, updates, etc.

What is Cybersecurity?

The body of technologies, processes and practices designed to protect networks, computers, programs, digital assets, and data (processed, stored and/or in transit) from attack, damage or unauthorized access.



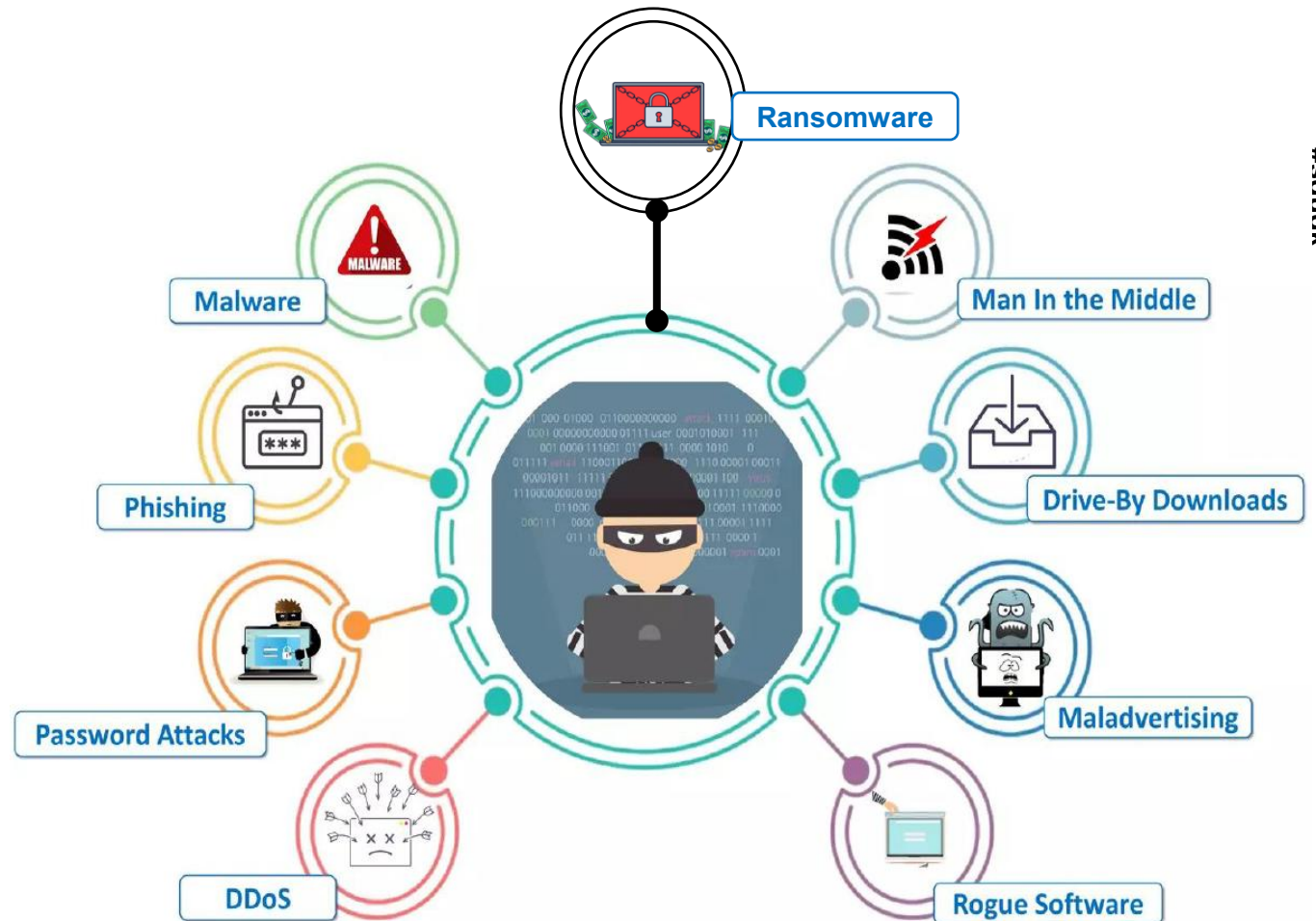
What is Cybersecurity?

“Measures and controls that ensure confidentiality, integrity, and availability of information system assets including hardware, software, firmware, and information being processed, stored, and communicated.”

[NIST Internal/Interagency Report NISTIR 7298 Rev. 3 (Glossary of Key Information Security Terms , July 2019)]

Why Cybersecurity?

- Ever increasing information sharing.
- Ever increasing interconnectedness.
- Ever increasing vulnerabilities & exploits.



Ever increasing cyber attacks

What to Protect – Against What?

Networks



All networks that interconnects our digital assets.

Systems and Devices



All systems and devices connected to the networks

Programs



All programs or source codes that built up systems connected to the networks.

*And most importantly:
Data*



All data, whether in transit, at rest, or processed.

Protect Against: Unauthorized Access, Modification, and Deletion

How to Protect: the CIA Triad



Confidentiality



Confidentiality is about preventing the disclosure of information to unauthorized parties.



Integrity



Integrity is about protecting information from being modified by unauthorized parties.



Availability



Availability is making sure that authorized parties are able to access the information when needed

Security Goals: the CIA Triad

- **Confidentiality**

E.g., encryption, access control, authentication.

- **Integrity**

E.g., signatures, checksums.

- **Availability**

E.g., computational redundancies, backups, mitigation strategies.



Where to Begin?

Do the involved technologies implement effective security measures?

- **Networks:** Transmission security (secure protocols), access control, etc
- **Desktop devices:** OS fragmentation, timely security updates, licensed softwares.
- **Mobile devices:** OS fragmentation, outdated HW without security support, user devices in secure environment.
- **Storage:** Localized vs mobile, data encrypted, biometric, etc.

Attack Surface

Attack Surface

A sum of all points where an attacker could try to enter, interact with, or extract data from a system.

Consist of the reachable and exploitable vulnerabilities in a system

Examples:

Open ports on outward facing Web and other servers, and code listening on those ports

Services available on the inside of a firewall

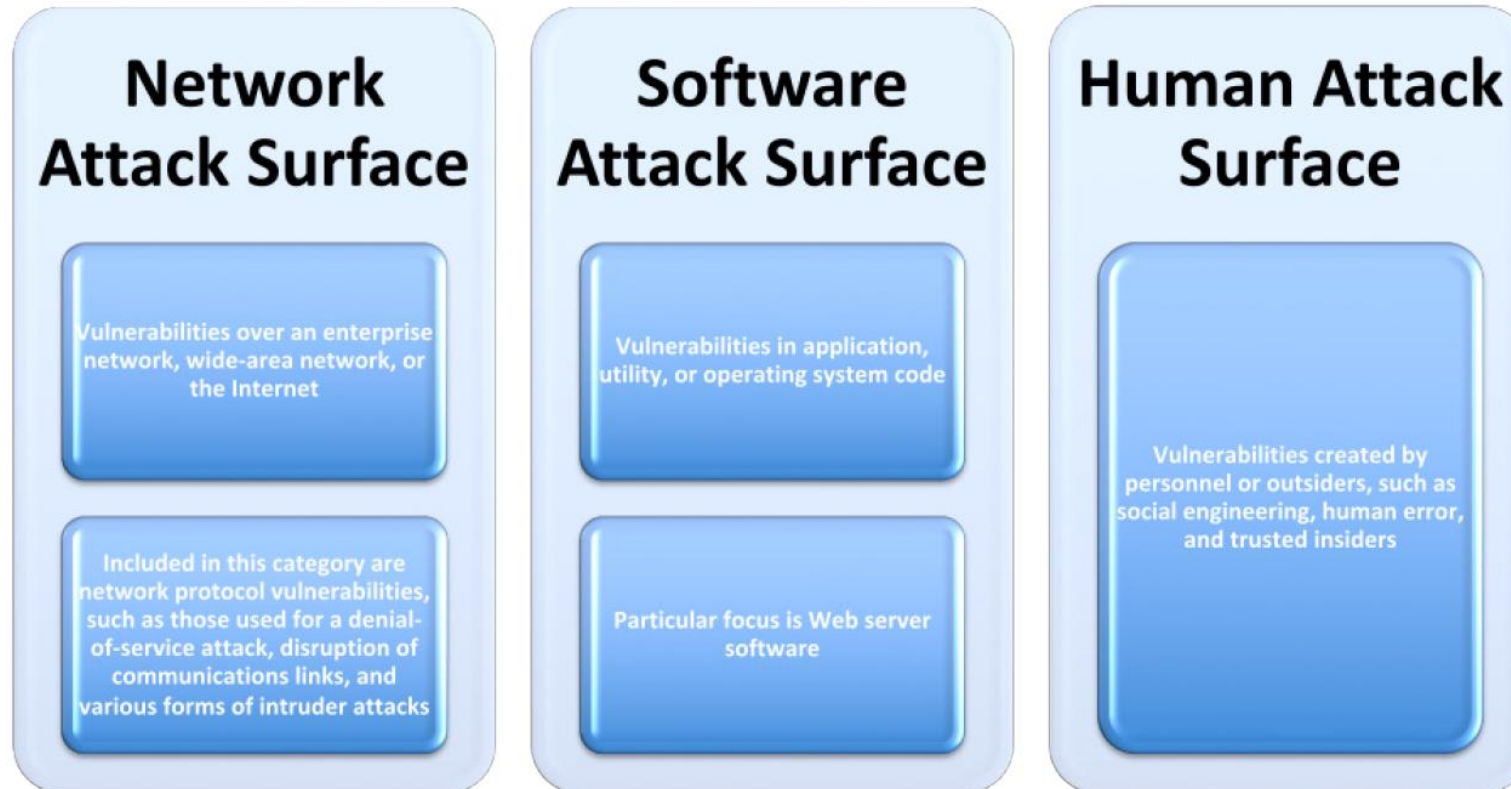
Code that processes incoming data, email, XML, office documents, and industry-specific custom data exchange formats

Interfaces, SQL, and Web forms

An employee with access to sensitive information vulnerable to a social engineering attack

Attack Surface

Attack vectors or end points which an attacker can use to negatively affect resources or data, e.g., an unauthorized user accessing/editing/deleting data.



Attack Surface

How can the data/system be accessed?

- Network
- Machine (physically)
- Software
- Human

⇒ **Attack Vectors**

Attack Surface = Σ of Attack Vectors

How to put the data into or
get it out of the system
securely?

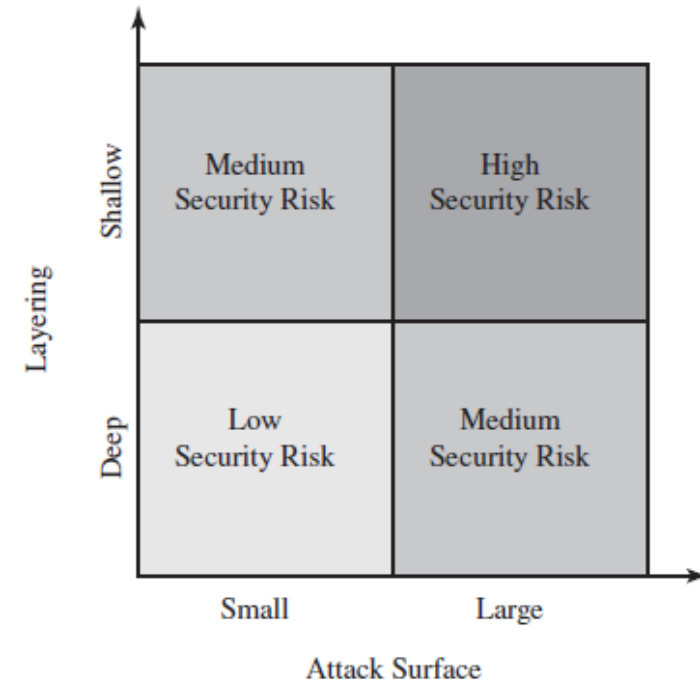
Attack Surface

Major entities of an attack surface

Attacker: Wants to expand the exposed interface.

Operator: Wants to limit the surface open to attacks.

Designer: Should design the system accordingly (operators' requirements).



Defense in Depth and Attack Surface

Access to the Target

- Depending on the kind of access, no vulnerability is needed (direct access)
- Social engineering is often a first step
- This can include an email to all / selected employees with or without crafted attachments.



Key Terms in Cybersecurity

- **Risk:** A measure of the extent to which an entity is threatened by a potential circumstance/event.
- **Vulnerability:** A weakness in a system
- **Threat:** Circumstance/event that have the potential to exploit a vulnerability
- **Exploit:** A method of taking advantage of a vulnerability
- **Attack:** An action that uses a threat to exploit a vulnerability
- **Patch:** A fix for a vulnerability
- **Countermeasure.** Means of preventing an attack, which tries to exploit one or more vulnerabilities

Key Takeaways

- Security is a wide field
- There is no silver bullet
- The process can be domain specific
- ... and iterative
- But there are procedures & best practices

Our focus:

- What do we want to protect?
- How to protect it safely?
- Where and how it can be attacked?
- How can these attacks be mitigated?
- What next when the worse case occurs?

Standards

Standards have been developed to cover management practices and the overall architecture of security mechanisms and services. The most important of these organizations are:

- National Institute of Standards and Technology (NIST)
- Internet Society (ISOC)
- International Telecommunication Union (ITU-T)
- International Organization for Standardization (ISO)

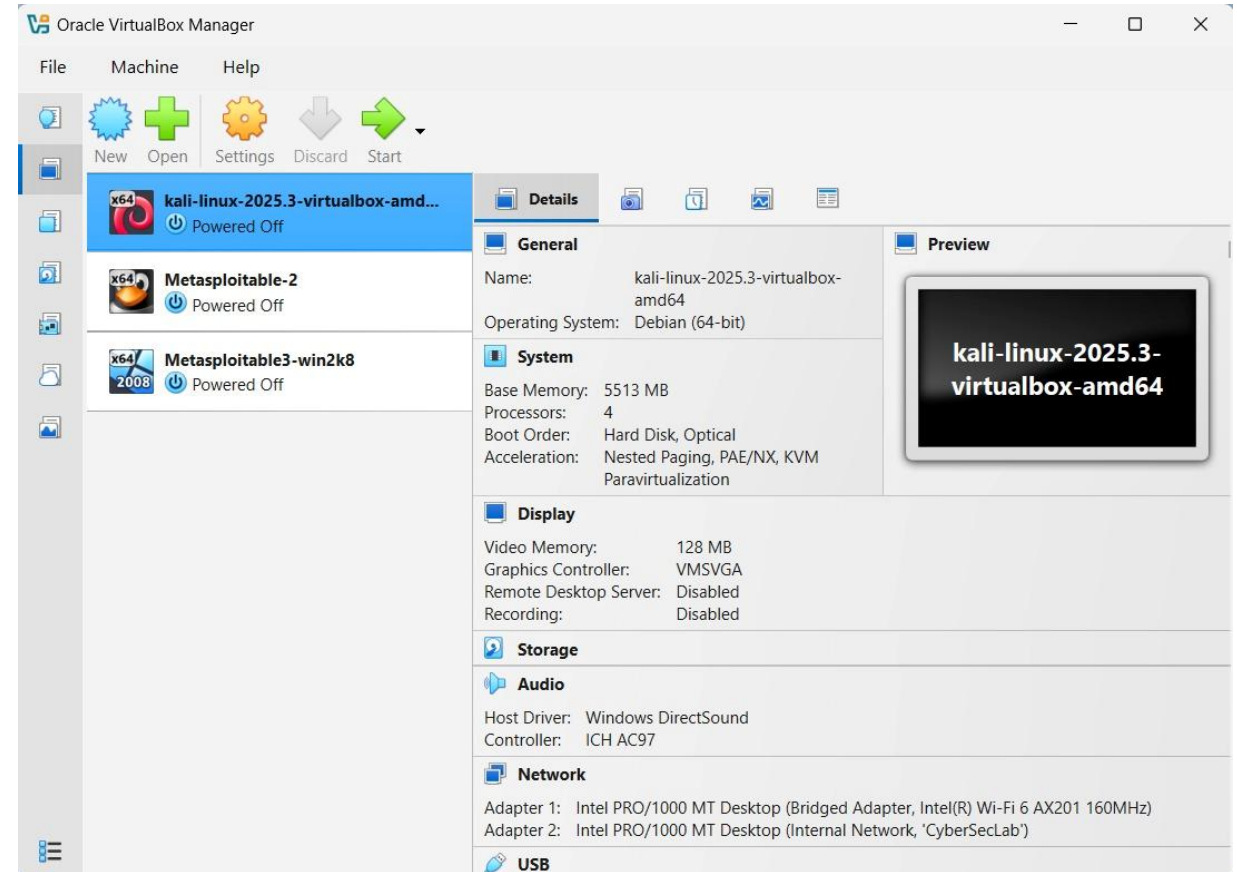
Legal Issues

- The Computer Security Act of 1987 & 1992
- OMB Circular A-130
- See www.alw.nih.gov/Security/FIRST/papers/legal/statelaw.txt for state computer laws
- Health Insurance Portability and Accountability Act of 1996, HIPAA
- EU Data Protection Regulations (GDPR 2018)

Lab Set-up

Lab Set-up for Exercises

- Attacker's / tester's machine: Kali Linux
- Victim / target machines:
 - ✓ Metasploitable 2 (Linux)
 - ✓ Metasploitable 3 (Windows)
- Everything is virtualized in VirtualBox



VirtualBox Networking

	VM ↔ Host	VM1 ↔ VM2	VM → Internet	VM ← Internet
HostOnly	Yes	Yes	No	No
Internal	No	Yes	No	No
Bridged	Yes	Yes	Yes	Yes
NAT	No	No	Yes	Port forwarding
NAT Network	No	Yes	Yes	Port forwarding

Host Only

- Limit VM to local private network
- Communicate with host or other VMs
- Neither send nor receive from Internet

Bridged Network

- VM is a node in the local network
- Pkg sniffing on host interface possible

Internal Networking

- VMs assigned to virtual private networks.

Network Address Translation (NAT)

- VM hidden by host
- VMs cannot communicate
- Traffic from VM appears as if from the host

NAT Network

- Assign VMs to virtual private networks
- Assigned VMs can communicate
- Traffic from VMs appears as if from the host

Kali Linux

- Linux Distribution based on Debian
- Designed for Penetration testing
- Open source project
- Maintained and funded by Offensive Security
- Download VirtualBox image for Kali (<https://kali.org>)



[Kali Linux](https://kali.org)

Kali Linux: Login and Config

- Since release 2020.1, Kali adheres to a non-root user policy.

Username: kali

Password: kali (I'll recommend you change the password after set-up)

- Use `sudo [-i]` to become or act as root.
- Permanently reconfigure the keyboard
\$ `dpkg-reconfigure keyboard-configuration`
- Temporarily switch the GUI's keyboard layout at runtime
\$ `setxkbmap en_UK`
\$ `setxkbmap dk`

Kali Linux: Update

- Update your Kali machine regularly.
- On the command line, use (as root or with sudo)
 - \$ apt update
 - \$ apt full-upgrade
 - Additionally, when suggested:
 - \$ apt autoremove
- Ofcourse, keep your vulnerable machines as outdated as they were designed.
- **Note:** Kali is not intended to be used as a desktop OS, but as a pen testing tool box.

Linux: Basic Commands

- Package manager – apt e.g., apt upgrade
- pwd – show location of current directory
- touch file – create file
- vi file – low bandwidth text editor, use vim for increased comfort
- cp source destination – copy file
- mv source destination – move file
- rm file – remove files
- rm -r directory – recursively removes directory and sub-directories
- grep string file – looks for occurrences of a string in one or more files
- strings – extract strings from binary file
- ip address, ip addr, or ip a – returns ip address

[More Basic Commands & Scripts](#)

Thank You!